

Unit 0: What are Differential Equations

Prepared by Staples Education

Practice Problems and Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They are graded from reading and classifying an equation, through verifying candidate solutions and applying initial conditions, up to two full physical models. Complete solutions follow in Part III.

1. Verify that $y = e^{3x}$ is a solution of $\frac{dy}{dx} = 3y$.
 2. State the order of each equation: (a) $\frac{dy}{dx} + y^2 = x$, (b) $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$, (c) $\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^5 = \sin x$.
 3. Classify each equation as linear or nonlinear, and name the offending term when it is nonlinear: (a) $\frac{dy}{dx} + x^2y = e^x$, (b) $\frac{dy}{dx} = y^2$, (c) $\frac{d^2y}{dx^2} + \sin y = 0$.
 4. The general solution of $\frac{dy}{dx} = 2x$ is $y = x^2 + C$. Verify it, then find the particular solution satisfying $y(1) = 5$.
 5. Show that $y = Ce^{-2x}$ satisfies $\frac{dy}{dx} + 2y = 0$ for every constant C , then find the C for which $y(0) = 4$.
 6. Verify that $y = \cos 2x$ is a solution of the second-order equation $\frac{d^2y}{dx^2} + 4y = 0$.
 7. Classify each as an ordinary or a partial differential equation: (a) $\frac{dy}{dt} = ky$, (b) $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, (c) $\frac{d^2x}{dt^2} + x = 0$.
 8. The general solution of $\frac{dy}{dx} = x + y$ is $y = Ce^x - x - 1$. Verify it, then find the particular solution with $y(0) = 2$.
 9. **(Application)** A population grows at a rate proportional to its size, $\frac{dP}{dt} = kP$, with $P(0) = 500$. If the population doubles in 4 hours, find k and write $P(t)$ explicitly.
 10. **(Application)** A radioactive sample obeys $\frac{dN}{dt} = -\lambda N$ with $N(0) = N_0$. Verify that $N(t) = N_0e^{-\lambda t}$ is a solution, then find λ given a half-life of 10 years.
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Part III — Complete Solutions

Solution 1. Differentiate the candidate and compare both sides.

$$\frac{dy}{dx} = 3e^{3x} \quad \text{and} \quad 3y = 3e^{3x}.$$

Both sides equal $3e^{3x}$ for all x , so $y = e^{3x}$ satisfies the equation.

Solution 2. The order is the highest derivative present.

(a) first order, (b) second order, (c) third order.

In (c) the fifth *power* on $\frac{dy}{dx}$ is a degree, not an order — the highest derivative is still the third.

Solution 3. An equation is linear when y and its derivatives appear only to the first power and are never composed inside another function.

(a) linear, (b) nonlinear: the term y^2 , (c) nonlinear: the term $\sin y$.

Solution 4. Differentiate to verify, then apply the condition.

$$\frac{d}{dx}(x^2 + C) = 2x \quad \checkmark$$

Apply $y(1) = 5$:

$$5 = (1)^2 + C \implies C = 4.$$

The particular solution is $y = x^2 + 4$.

Solution 5. Differentiate $y = Ce^{-2x}$ and substitute.

$$\frac{dy}{dx} = -2Ce^{-2x}$$

$$\frac{dy}{dx} + 2y = -2Ce^{-2x} + 2Ce^{-2x} = 0 \quad \checkmark$$

The equation holds for every C . Apply $y(0) = 4$:

$$y(0) = C = 4 \implies y = 4e^{-2x}.$$

Solution 6. Differentiate twice and substitute.

$$\frac{dy}{dx} = -2 \sin 2x, \quad \frac{d^2y}{dx^2} = -4 \cos 2x$$

$$\frac{d^2y}{dx^2} + 4y = -4 \cos 2x + 4 \cos 2x = 0 \quad \checkmark$$

So $y = \cos 2x$ is a solution of the second-order equation.

Solution 7. An ordinary equation has a single independent variable; a partial equation has several.

(a) ODE (only t), (b) PDE (t and x), (c) ODE (only t).

Solution 8. Differentiate $y = Ce^x - x - 1$ and check against $x + y$.

$$\frac{dy}{dx} = Ce^x - 1$$

$$x + y = x + (Ce^x - x - 1) = Ce^x - 1 \quad \checkmark$$

The two agree, so the family solves the equation. Apply $y(0) = 2$:

$$2 = Ce^0 - 0 - 1 = C - 1 \implies C = 3$$

$$y = 3e^x - x - 1.$$

Solution 9. A rate proportional to size gives exponential growth, $P(t) = P_0 e^{kt}$, and here $P_0 = 500$:

$$P(t) = 500 e^{kt}.$$

Doubling in 4 hours means $P(4) = 1000$:

$$1000 = 500 e^{4k} \implies e^{4k} = 2 \implies 4k = \ln 2$$

$$k = \frac{\ln 2}{4}.$$

Hence

$$P(t) = 500 e^{(\ln 2/4)t} = 500 \cdot 2^{t/4}.$$

Solution 10. First verify the candidate by differentiating:

$$\frac{dN}{dt} = -\lambda N_0 e^{-\lambda t} = -\lambda N \quad \checkmark$$

so $N(t) = N_0 e^{-\lambda t}$ solves the decay law. A half-life of 10 years means $N(10) = \frac{1}{2}N_0$:

$$\frac{1}{2}N_0 = N_0 e^{-10\lambda} \implies e^{-10\lambda} = \frac{1}{2}$$

$$-10\lambda = -\ln 2 \implies \lambda = \frac{\ln 2}{10} \text{ per year.}$$

Unit 0 Key Takeaway

Every problem here rests on one habit — treat the unknown as a **function** and let the equation test it. You **verify** a candidate by substituting it and its derivatives and checking both sides agree for all x ; you pass from the **general solution** to a particular curve by using an **initial condition** to fix the arbitrary constant. The same proportional rate law drives growth and decay alike.